HTTP

Hypertext Transfer Protocol

The World Wide Web

Is an information space
Where document and information resources
Are identified by Uniform Resource Locators (URLs),
Interlinked by hypertext links,
And can be accessed via the internet.

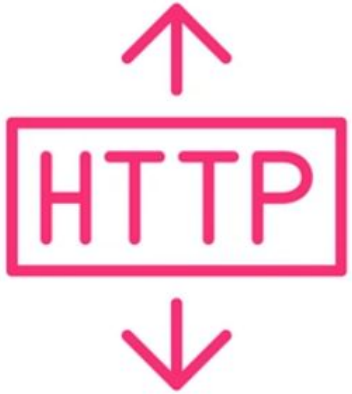
Ingredients of the Web



Resources (documents, images, other files)

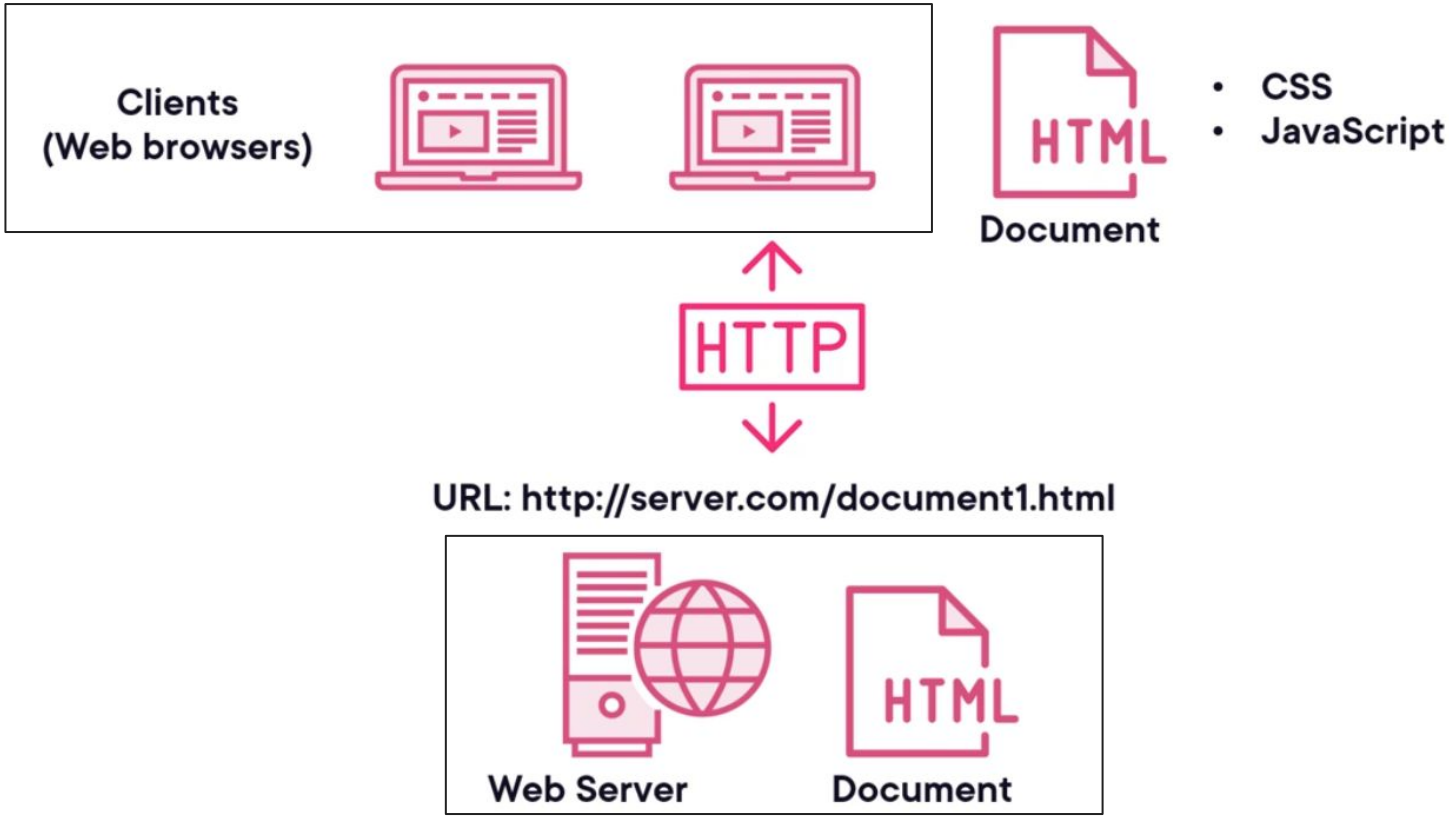


Uniform Resource Locators (URLs)



Hypertext Transfer Protocol (HTTP)

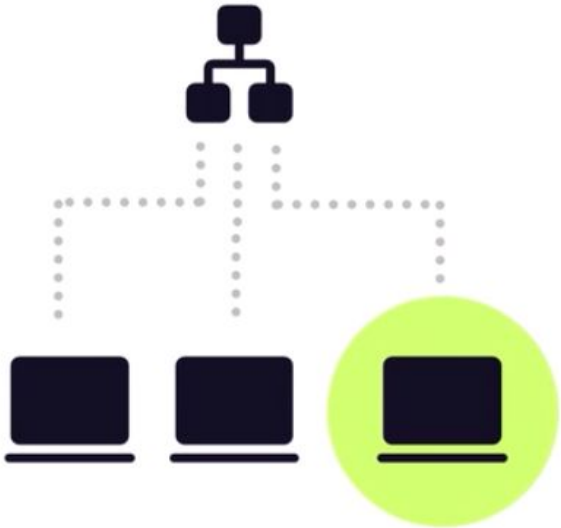
Ingredients of the Web

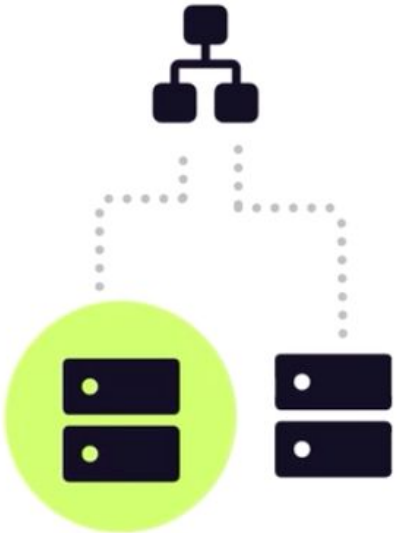


Accessing a Website

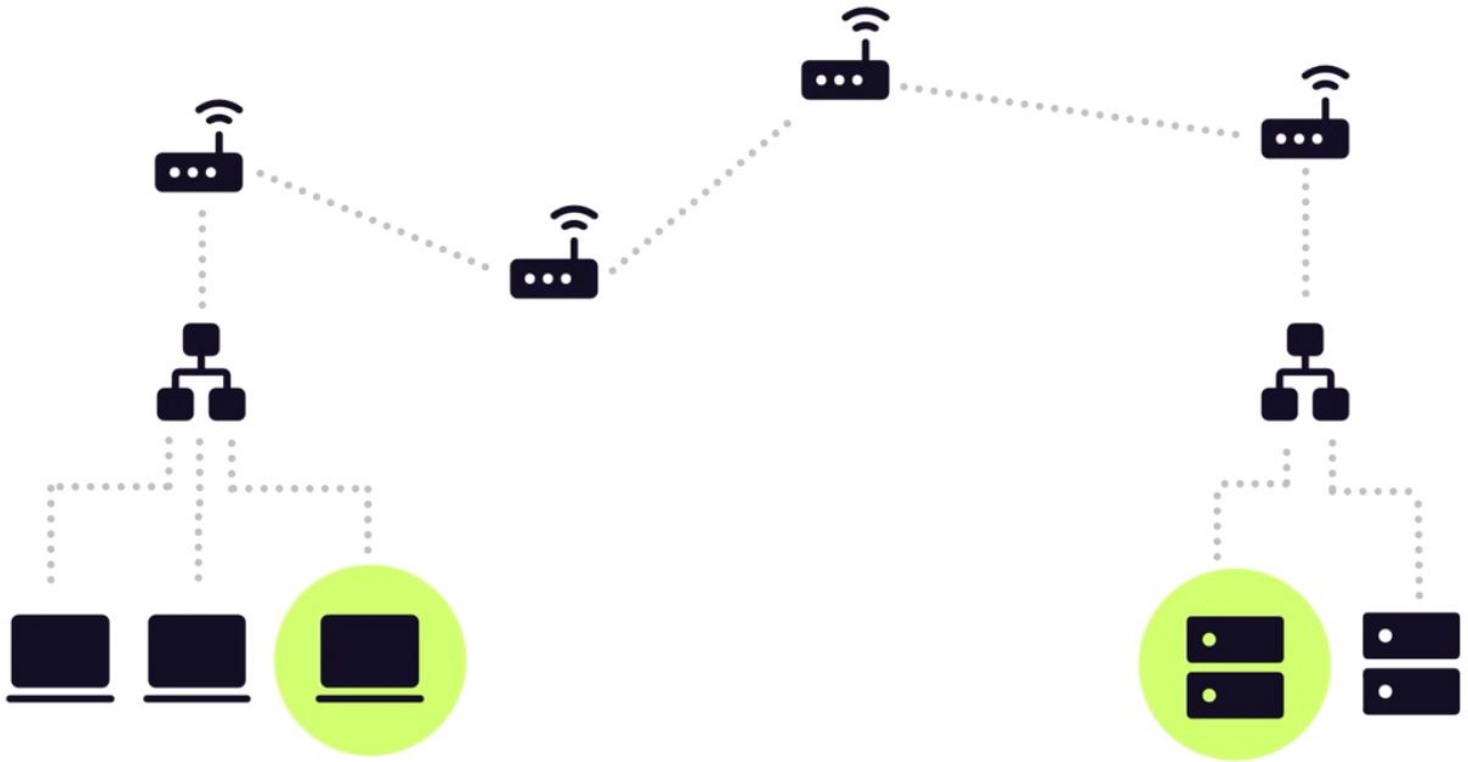


Accessing a Website

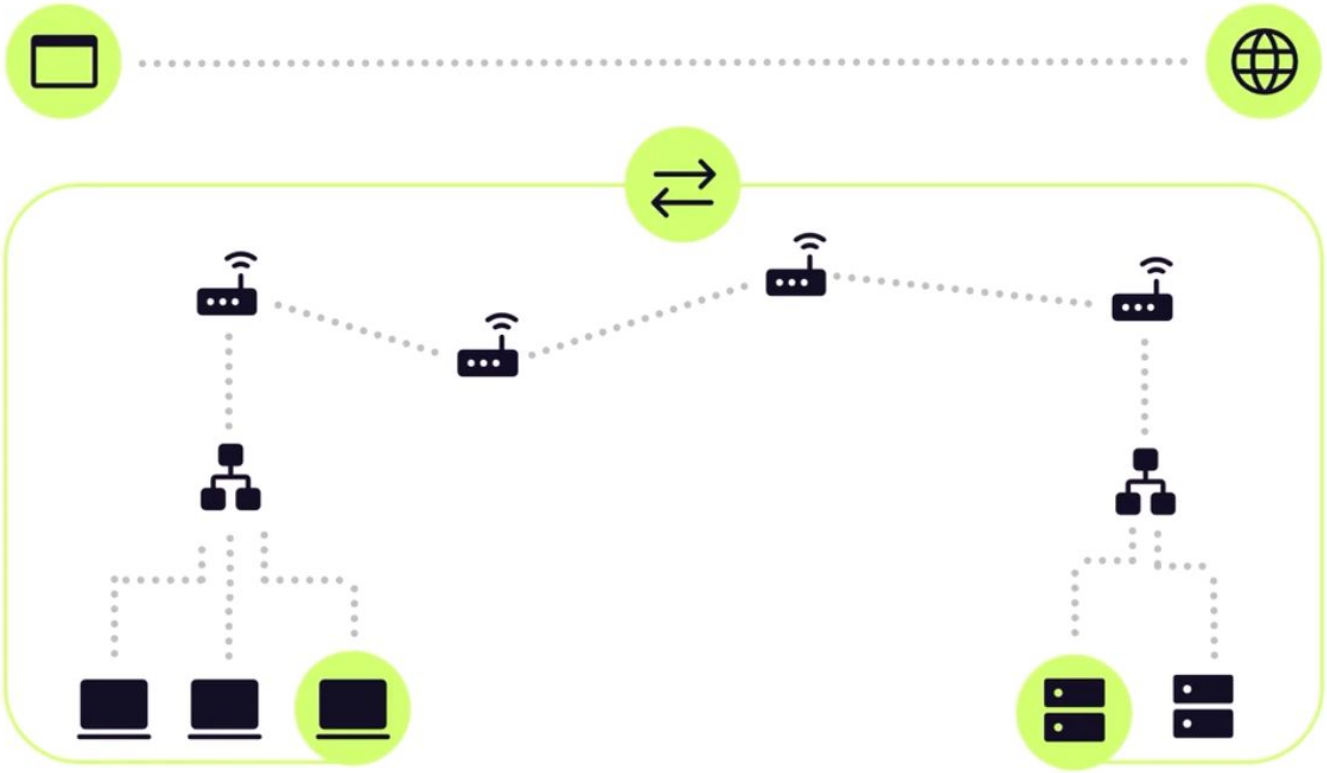




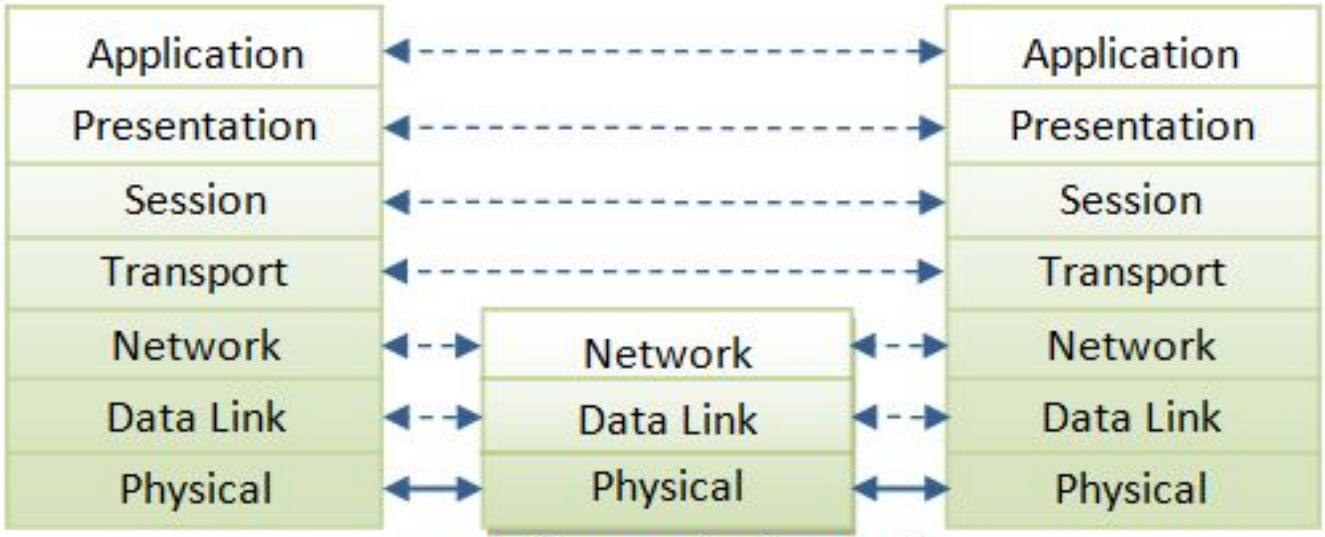
Accessing a Website

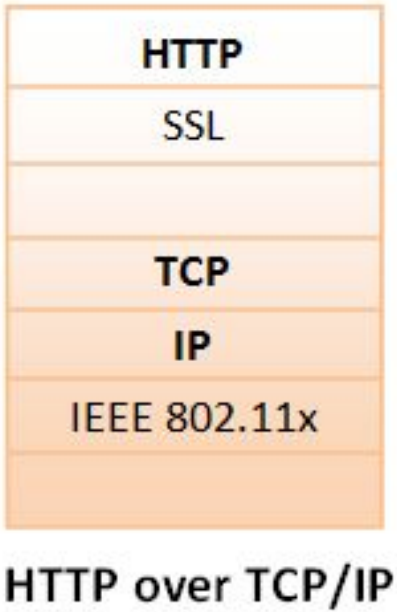


Accessing a Website



ISO OSI 7-layer network





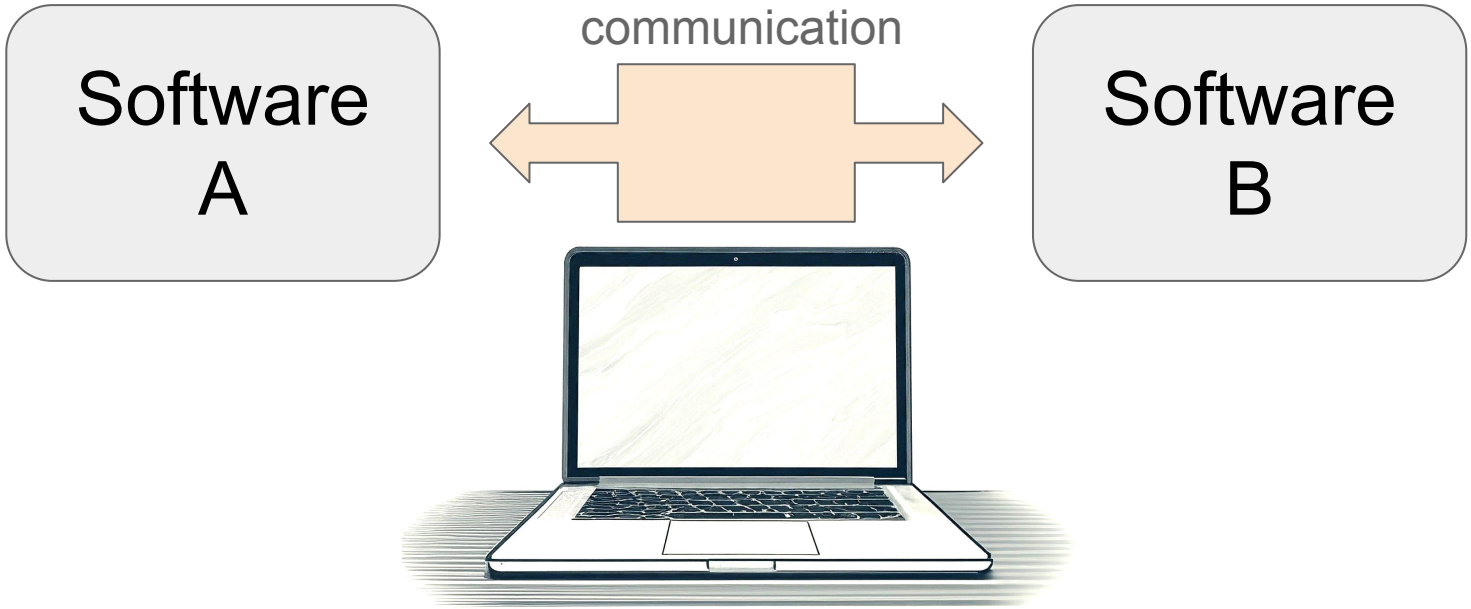
HTTP is ...

- a protocol for fetching resources such as
  - HTML documents
  - also images and videos
- a text-based client-server protocol
- an application layer protocol that is
  - sent over TCP

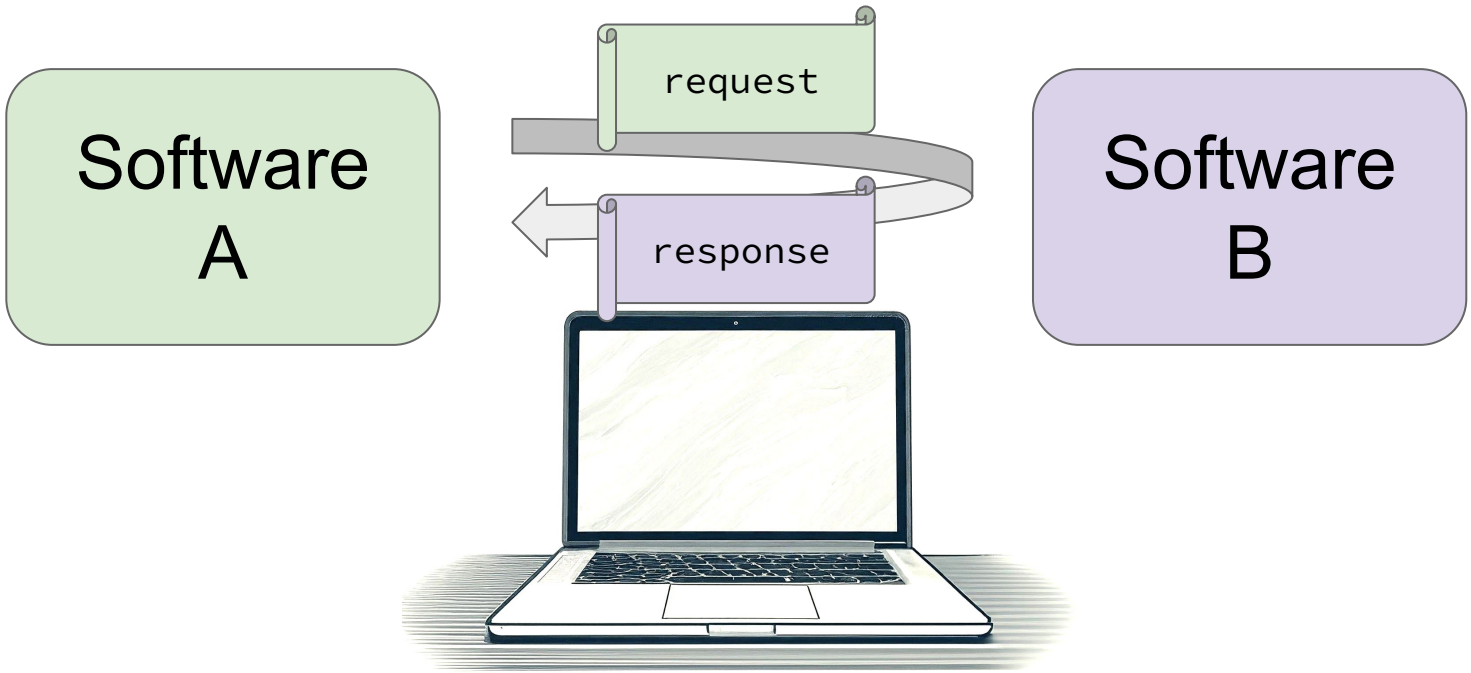
PORT

the default port for the HTTP protocol is 80

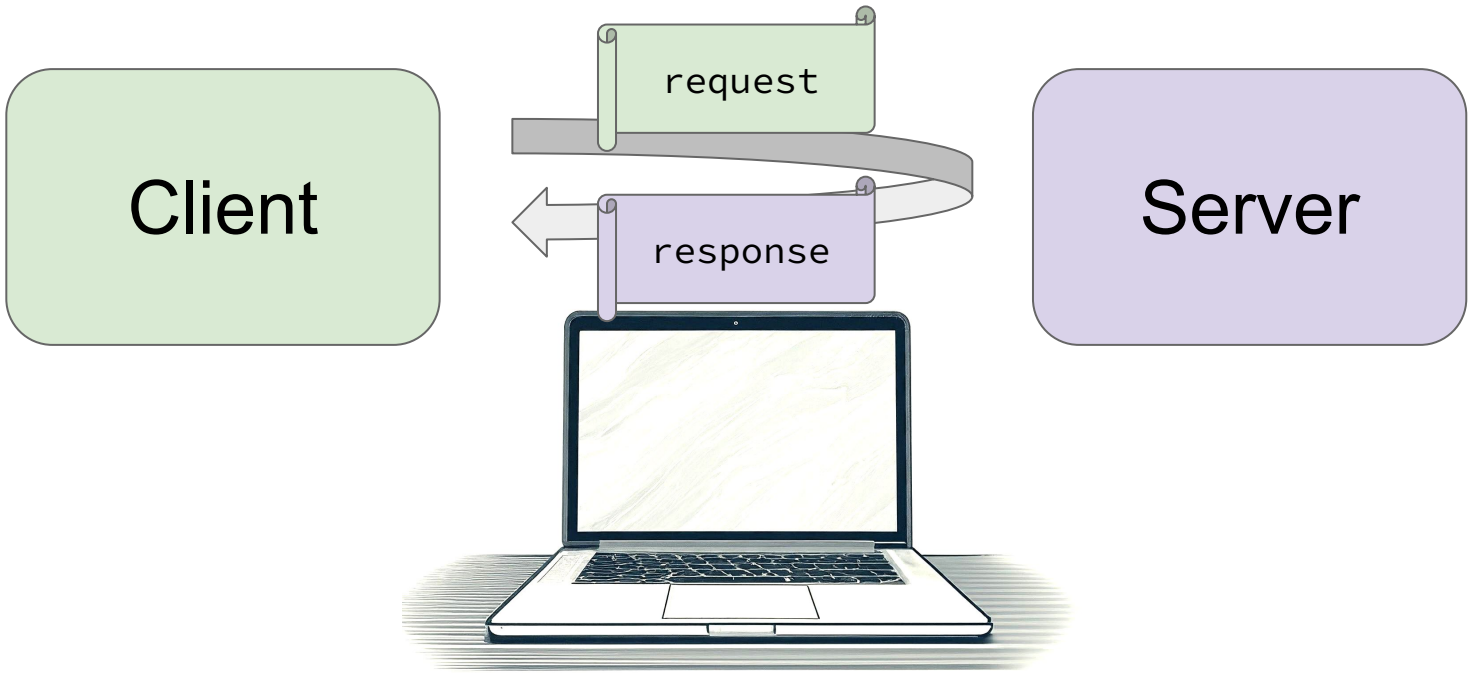
the default port for the HTTPS protocol is 443

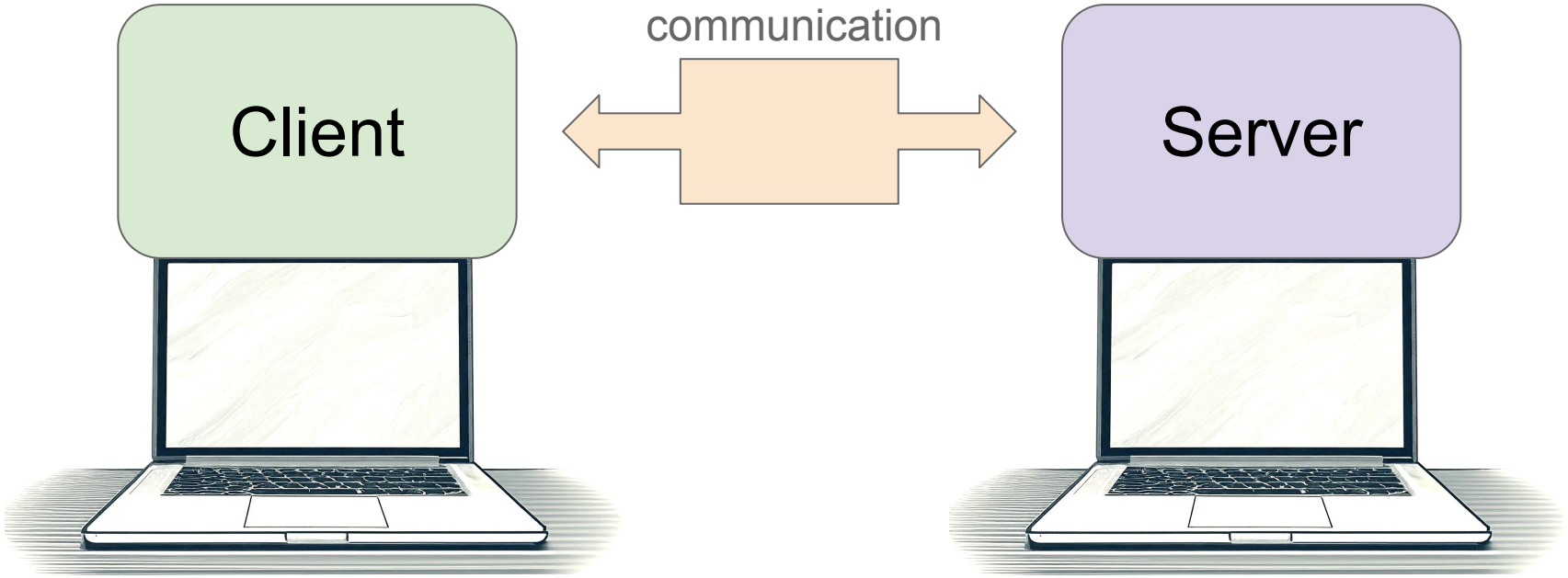

Software
A
communication
Software
B

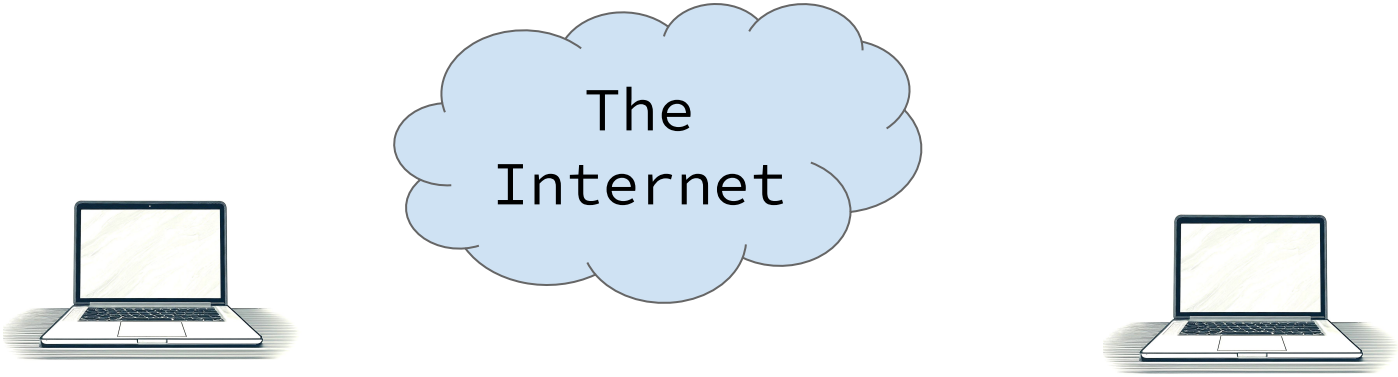
communication

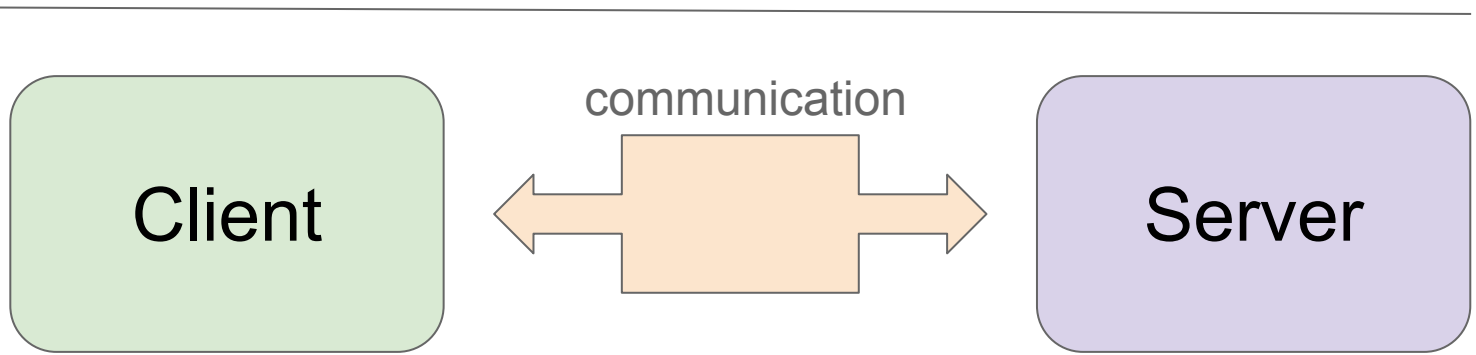


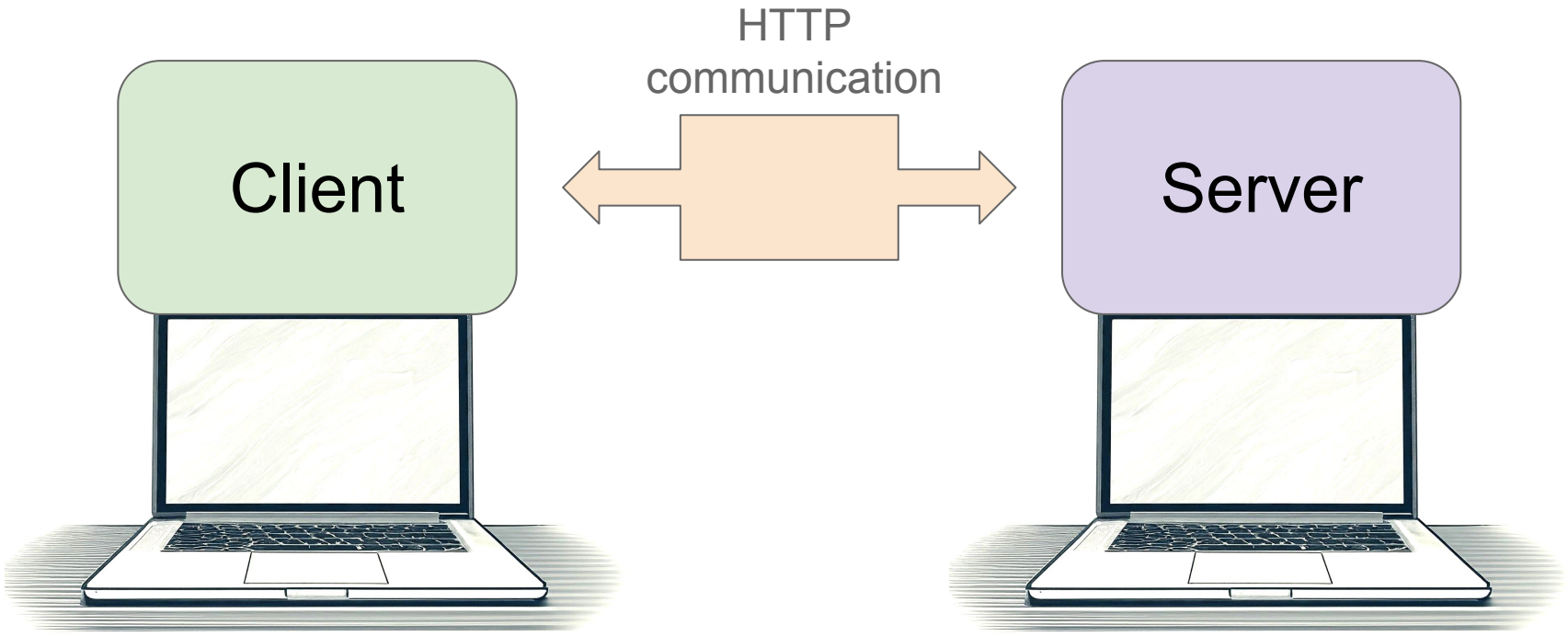
communication

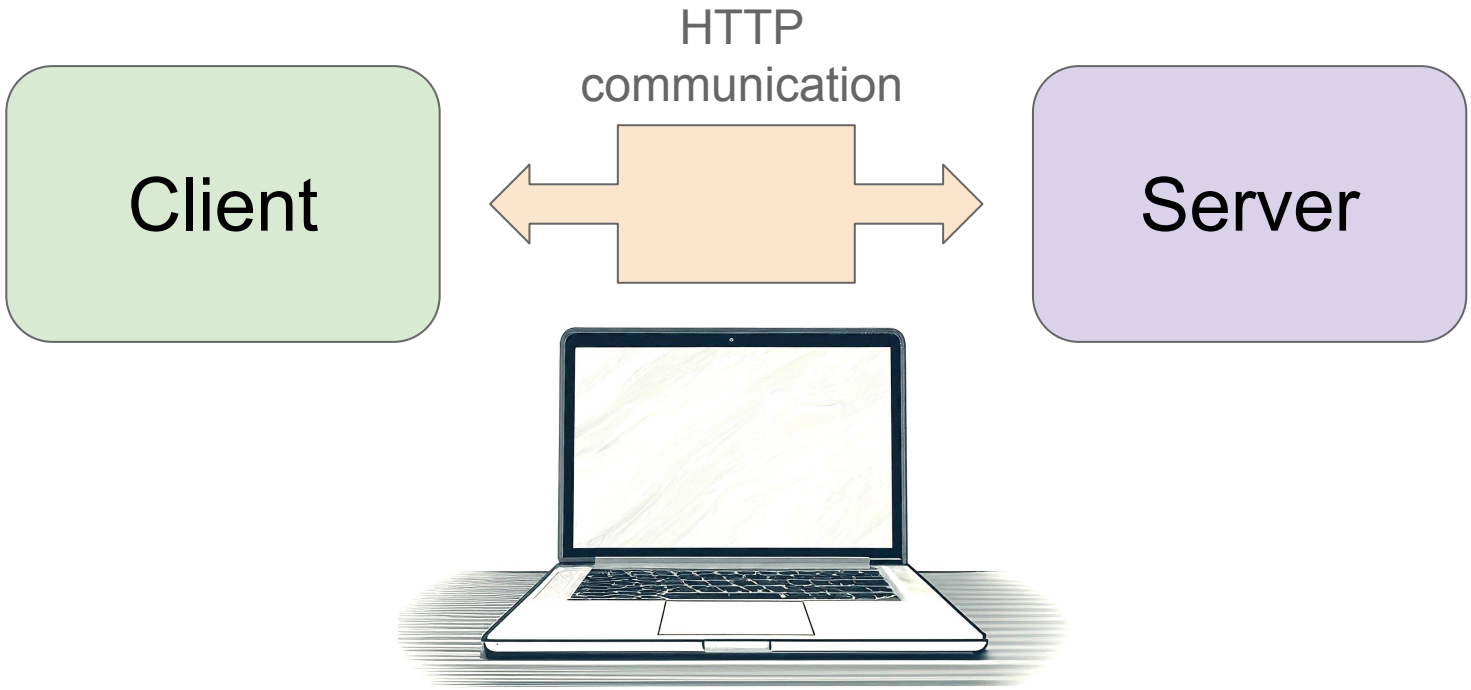



communication
Client
Server


The
Internet


communication
Client
Server


HTTP
communication
Client
Server


HTTP
communication
Client
Server

HTTP Flow

A client wants to communicate with a server, it performs the following steps:

1. Open a TCP connection
2. Send an HTTP message (request message)
3. Read the response sent back by the server
4. Close the connection
    - or reuse the connection for further requests

HTTP Conversation: Example

```
GET /trainings/courses HTTP/1.1
Host: softuni.org
User-Agent: Mozilla/5.0
<CRLF>
```

HTTP Request

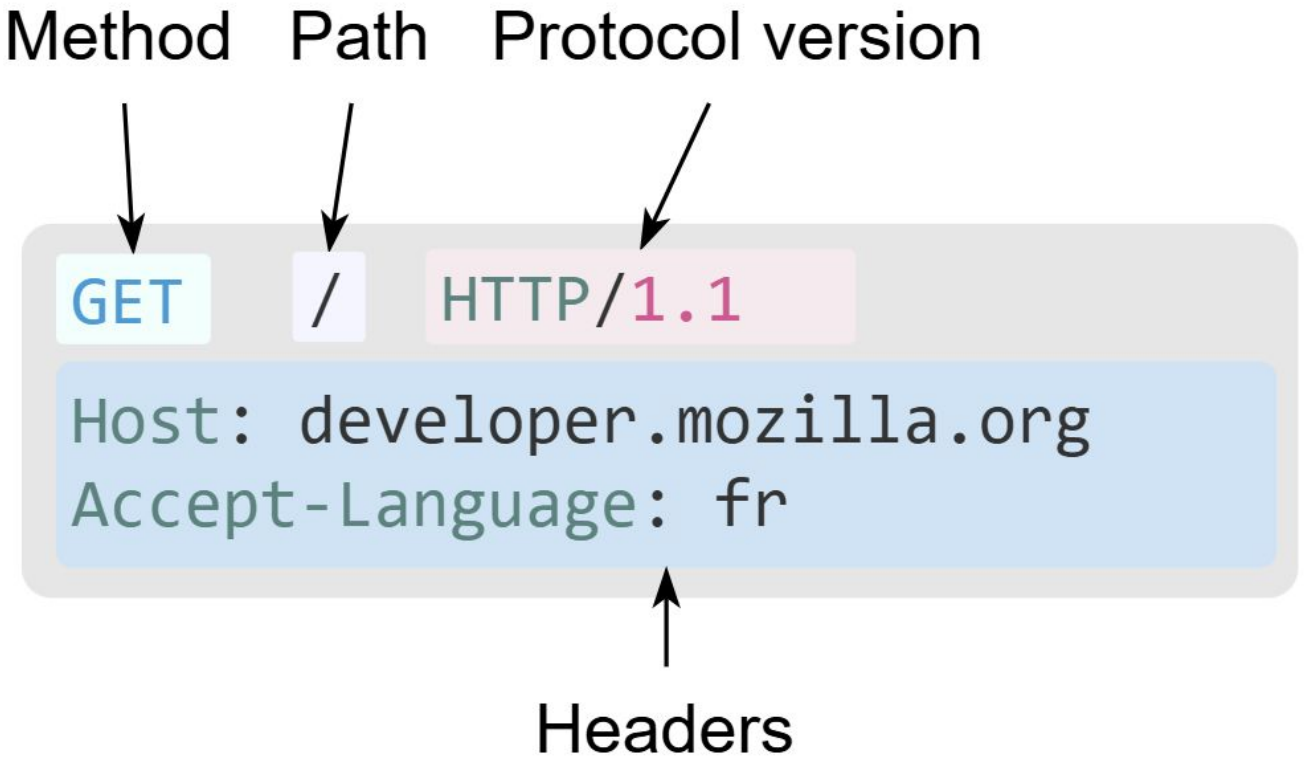
```
HTTP/1.1 200 OK
Date: Tue, 16 May 2020 15:13:41 GMT
Server: Microsoft-HTTPAPI/2.0
Last-Modified: Tue, 16 Jan 2018 15:13:42 GMT
Content-Length: 18586
<CRLF>
<html><title>Get a Tech Degree from…
</title>
```

HTTP Response

HTTP Request Message

```
GET / HTTP/1.1
Host: developer.mozilla.org
Accept-Language: fr
<crlf>
```

HTTP Request Message - Explained

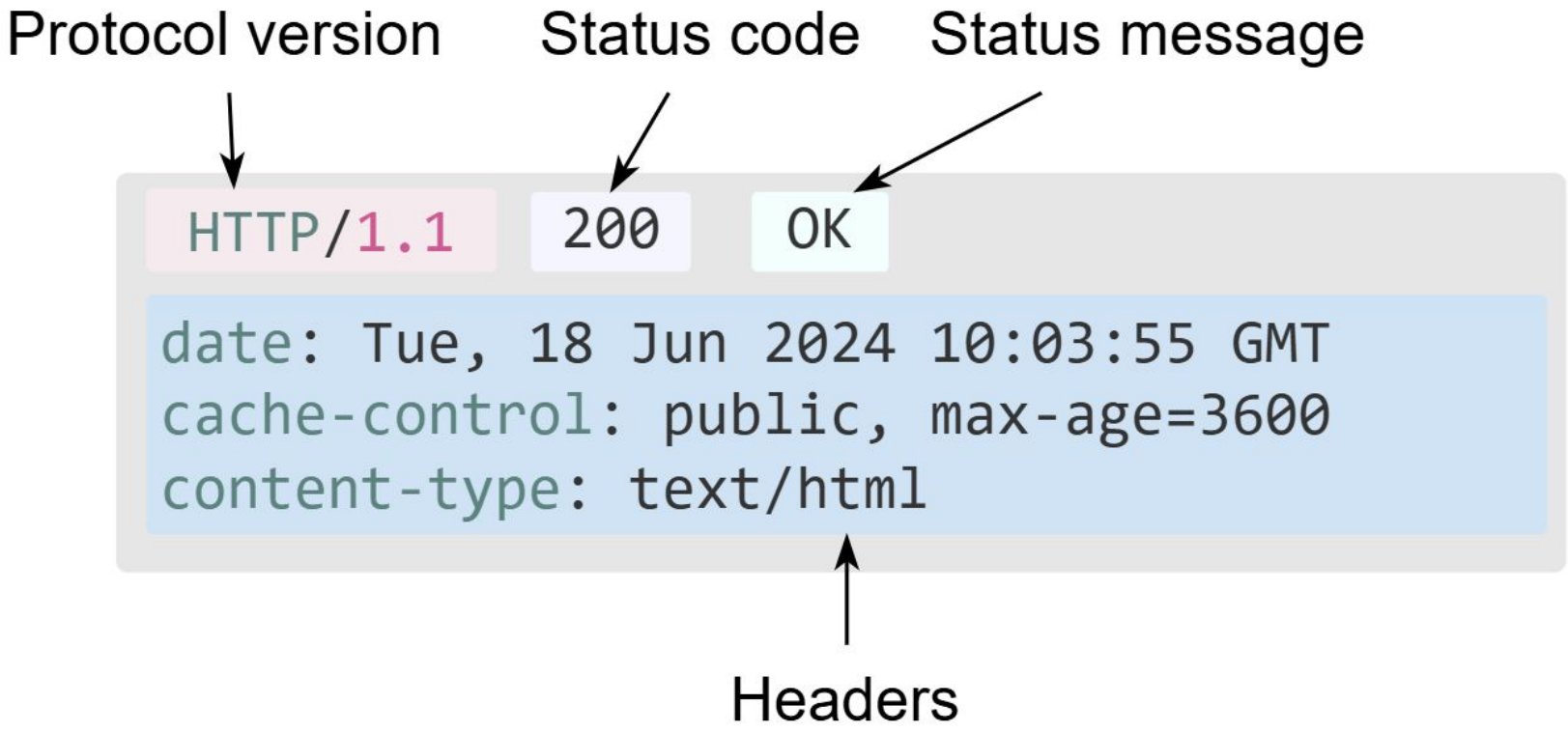


HTTP Response Message

```
HTTP/1.1 200 OK
Date: Sat, 09 Oct 2010 14:28:02 GMT
Server: Apache
Last-Modified: Tue, 01 Dec 2009 20:18:22 GMT
ETag: "51142bc1-7449-479b075b2891b"
Accept-Ranges: bytes
Content-Length: 29769
Content-Type: text/html

<!doctype html>… (here come the 29769 bytes of the requested web page)
```

HTTP Response Message - Explained



Response Status Codes

- 2xx for Success, the most common one is 200 which means “OK”.
- 3xx for Redirection, the most common ones are 301 and 303 which mean “Permanent Redirect” and “Redirect for Undefined Reason”, respectively.
- 4xx for Application Error, the most common ones are 403 and 404 which mean “Forbidden” and “Not Found”, respectively.
- 5xx for Server Error, the most common one is 500 which means “Server Error”.

HTTP Response Status Codes

Status Code	Action	Description
200	OK	Successfully retrieved resource
201	Created	A new resource was created
204	No Content	Request has nothing to return
301 / 302	Moved	Moved to another location (redirect)
400	Bad Request	Invalid request / syntax error
401 / 403	Unauthorized	Authentication failed / Access denied
404	Not Found	Invalid resource was requested
409	Conflict	Conflict was detected, e.g. duplicated email
500 / 503	Server Error	Internal server error / Service unavailable

Content-Type and Disposition

The **Content-Type** / **Content-Disposition** headers specify how to process the HTTP request / response body

```
Content-Type: application/json
```

JSON-encoded data

```
Content-Type: text/html; charset=utf-8
```

UTF-8 encoded HTML page

```
Content-Type: application/pdf
Content-Disposition: attachment;
filename="Financial-Report-2020.pdf"
```

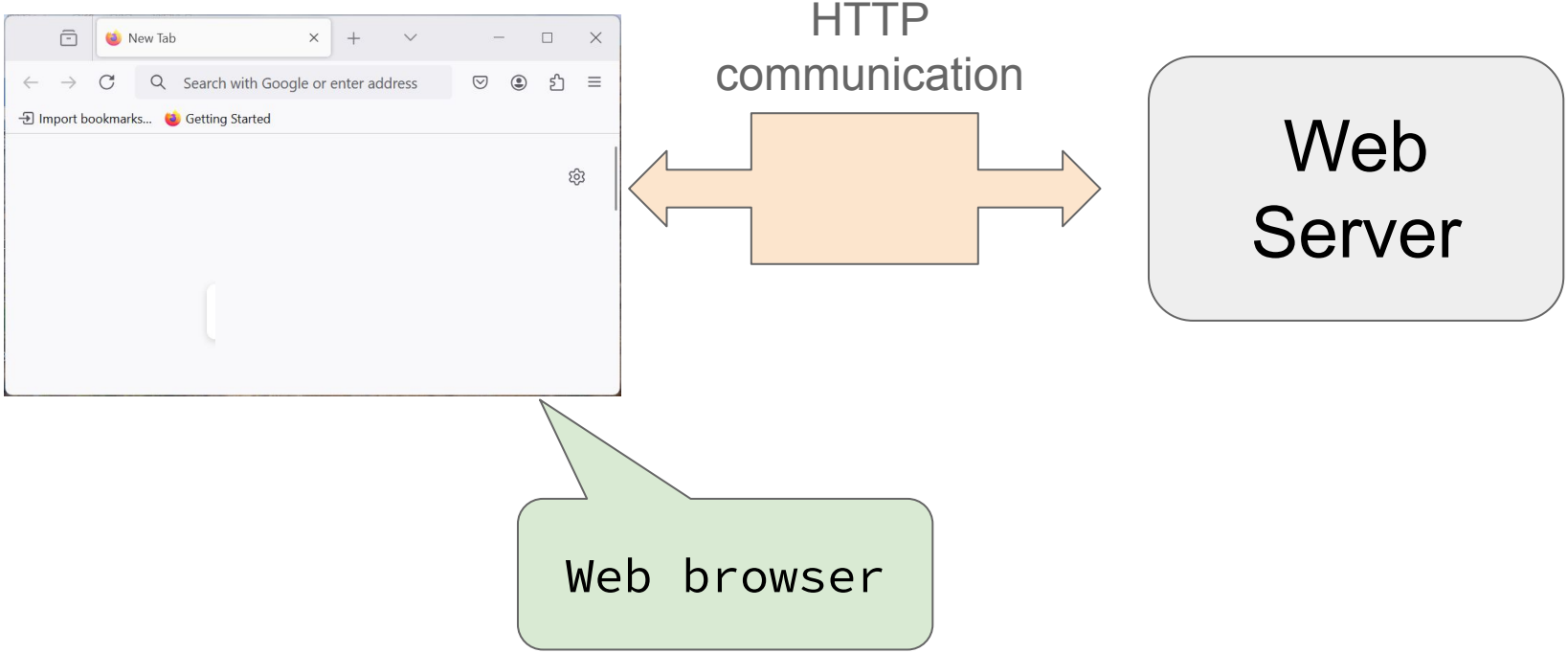
Download a PDF file

Standard media type: https://iana.org/assignments/media-types

Content-Type: Examples

- text/html
- text/css
- text/javascript
- application/javascript
- font/ttf
- font/otf
- video/mp4
- image/jpeg
- image/gif
- image/png
- image/svg+xml

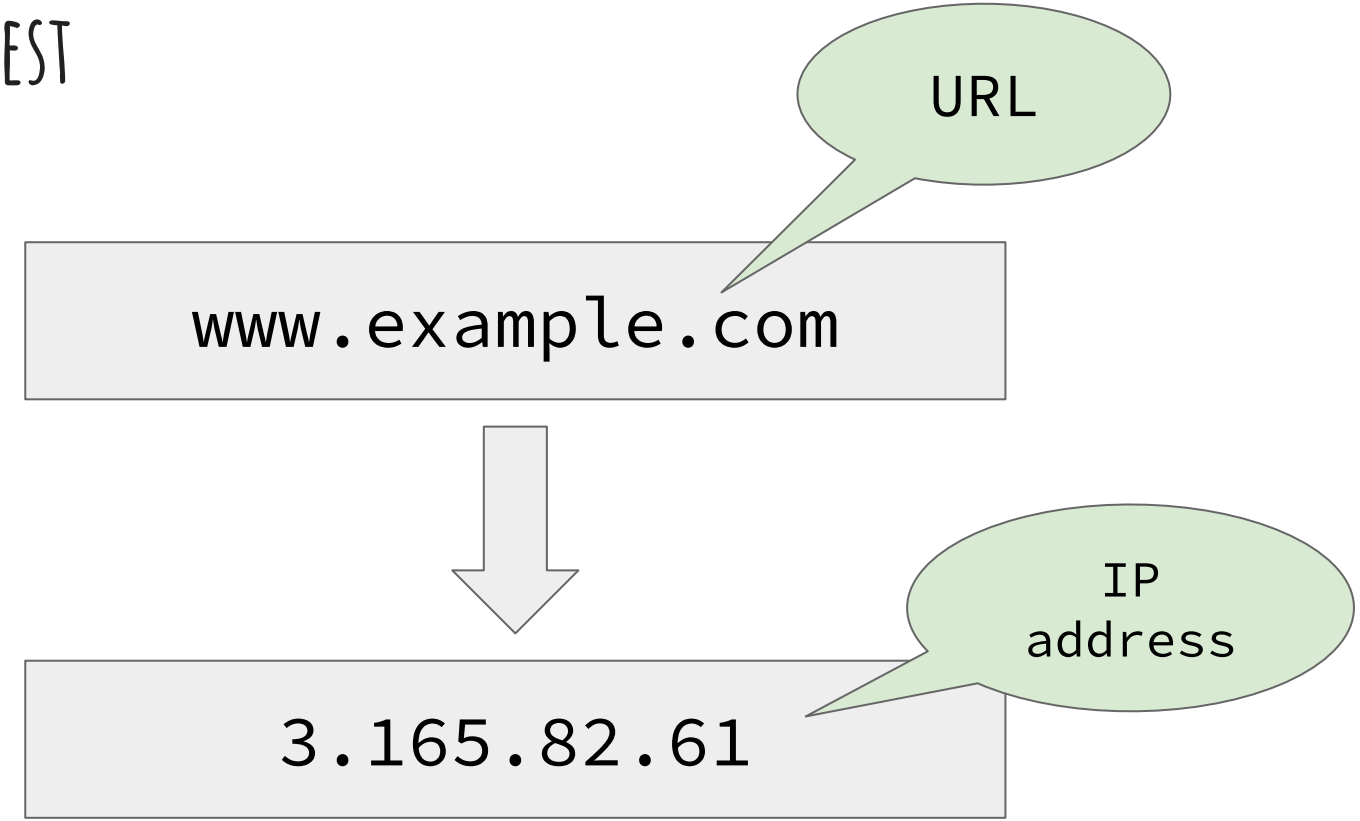
- multipart/form-data
- application/json
- application/yaml
- application/xml
- text/csv
- text/calendar


New Tab
Search with Google or enter address
Import bookmarks...
Getting Started
HTTP
communication
Web
Server
Web browser

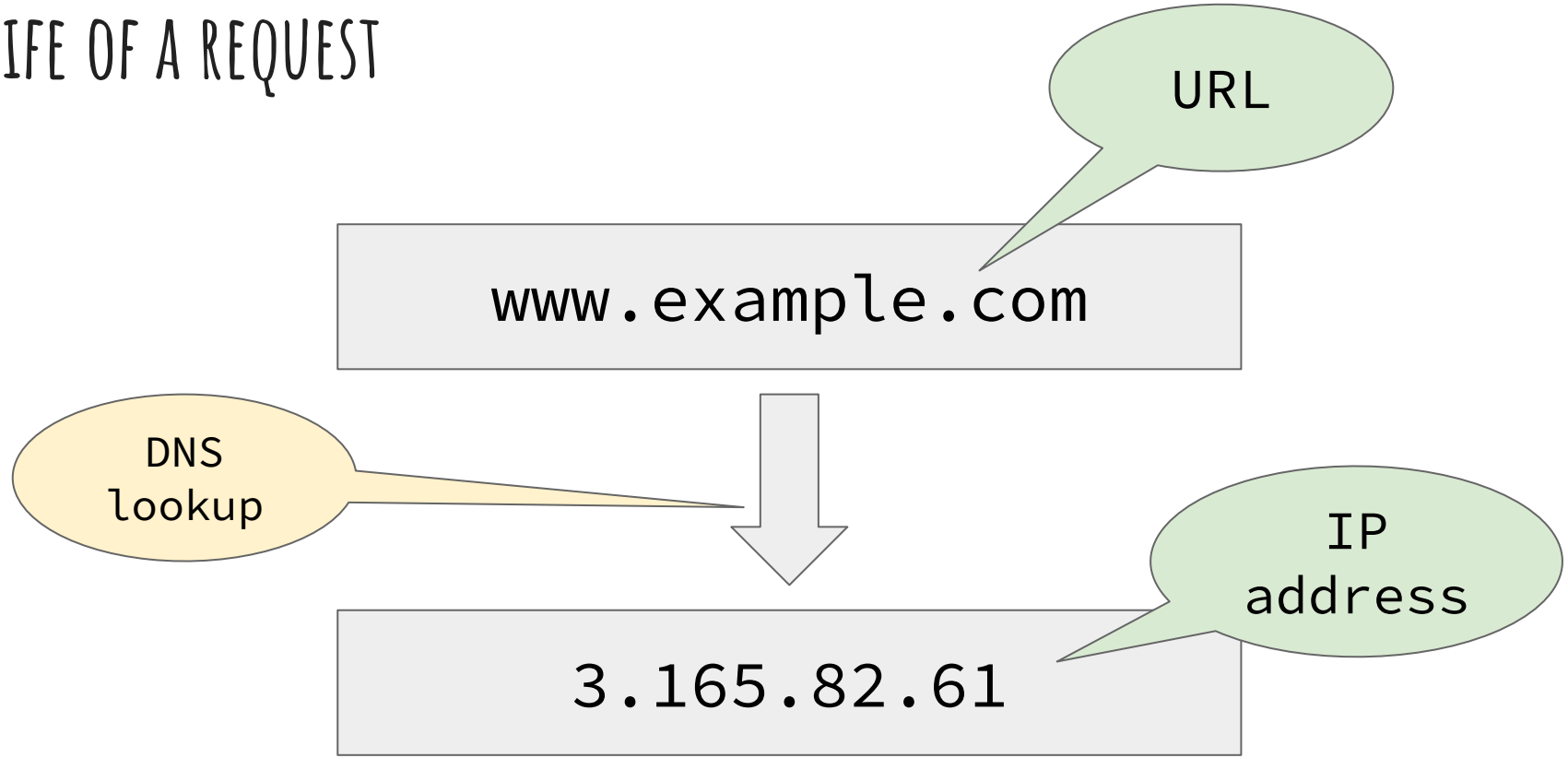
URL


BREAKDOWN OF A URL
PROTOCOL
DOMAIN
https://www.website.com/view/page-one
SUBDOMAIN
PATH & PAGE

Life of a request



Life of a request



What happens when you type google .com into a browser?

https://yt3.ggpht.com/PY_WVJQA73zmMTzP4WAavISV5KZA9yNYUJkDYTVaJVKMoWzakdGH0pgB3Z1B30_9PIPXKraT4Xy78S0=s1600-nd-v1-rwa

1. First up, you type the website address in the **browser's address bar**.
2. The **browser checks its cache** first. If there's a cache miss, it must find the IP address.
3. **DNS lookup** begins (think of it as looking up a phone number). The request goes through different DNS servers (root, TLD, and authoritative). Finally, the IP address is retrieved.
4. Your browser initiates a **TCP connection** like a handshake. For example, in the case of HTTP 1.1, the client and server perform a TCP three-way handshake with SYN, SYN-ACK, and ACK messages.
5. Once the handshake is successful, the browser **makes an HTTP request** to the server and the **server responds with HTML, CSS, and JS files**.
6. Finally, the browser processes everything. It **parses the HTML document** and **creates DOM and CSSOM trees**.
7. The browser **executes the JavaScript and renders the page** through various steps (tokenizer, parser, render tree, layout, and painting).
8. Finally, the **webpage appears on your screen**.

What is HTTPS?

HTTPS is a secure way to send data
between a web server and a web browser.

HTTP is stateless:

there is no link between two requests being successively carried out on the same connection.

HTTP is stateless, but not sessionless